

All rings are assumed commutative. The tensor product can be defined for non-commutative rings, but will in general just be an abelian group, not a module. Proceed with caution.

Definition: The tensor product of two R modules M and N is defined to be the R -module $M \otimes N$ generated by the simple tensors of the form $x \otimes y$ with $x \in M$, $y \in N$, subject to the relations

$$\begin{aligned}(x + x') \otimes y &= x \otimes y + x' \otimes y \\ x \otimes (y + y') &= x \otimes y + x \otimes y' \\ (rx) \otimes y &= r(x \otimes y) \\ x \otimes (ry) &= r(x \otimes y).\end{aligned}$$

Universal Property: Let M , N , and P be R -modules. An R -bilinear map $M \times N \rightarrow P$ is a function which is a linear map in either argument when the other argument is held constant. The map $\otimes : M \times N \rightarrow M \otimes N$ given by $\otimes(x, y) = x \otimes y$ is bilinear, and given any bilinear map $b : M \times N \rightarrow P$, there exists a unique R -module homomorphism f such that the diagram below commutes.

$$\begin{array}{ccc} M \times N & \xrightarrow{\otimes} & M \otimes N \\ & \searrow b & \downarrow f \\ & & P \end{array}$$

Slogans: The tensor product is commutative, associative, and distributes over direct sums. Also, tensoring an R -module with R doesn't do anything. In symbols:

$$\begin{aligned}M \otimes N &\cong N \otimes M \\ (M \otimes N) \otimes P &\cong M \otimes (N \otimes P) \\ (M \oplus N) \otimes P &\cong (M \otimes P) \oplus (N \otimes P) \\ R \otimes M &\cong M.\end{aligned}$$

Exactness: The tensor product is right-exact, that is, if

$$M' \xrightarrow{f} M \xrightarrow{g} M'' \rightarrow 0$$

is exact, then so is

$$M' \otimes N \xrightarrow{f \otimes 1} M \otimes N \xrightarrow{g \otimes 1} M'' \otimes N \rightarrow 0.$$

Vector Spaces: If V and W are finite dimensional vector spaces over a field k , then $\dim(V \otimes W) = (\dim V) \cdot (\dim W)$.

Extension of scalars: If M is an A module and B is an A algebra, then $B \otimes_A M$ has the structure of a B module with scalar multiplication given by $b(b' \otimes x) = bb' \otimes x$. The following rules are helpful for dealing with extensions of scalars:

$$\begin{aligned}B \otimes_A A[t] &\cong B[t] \\ B \otimes_A (A/I) &\cong B/I^e \\ (A/I) \otimes_A M &\cong M/IM \\ S^{-1}A \otimes_A M &\cong S^{-1}M.\end{aligned}$$

Tensor Algebras: If B and C are A -algebras, then $B \otimes_A C$ is given an A -algebra structure by defining multiplication by $(b \otimes c)(b' \otimes c') = bb' \otimes cc'$.